Certification Program: IBM Skillsbuild PBL- From Learner to become An AI Agent Architect

Project Title :

VocalEyes "Hear the world, see the possibilities."

Presented By: Gen-Z AI

Team Members:

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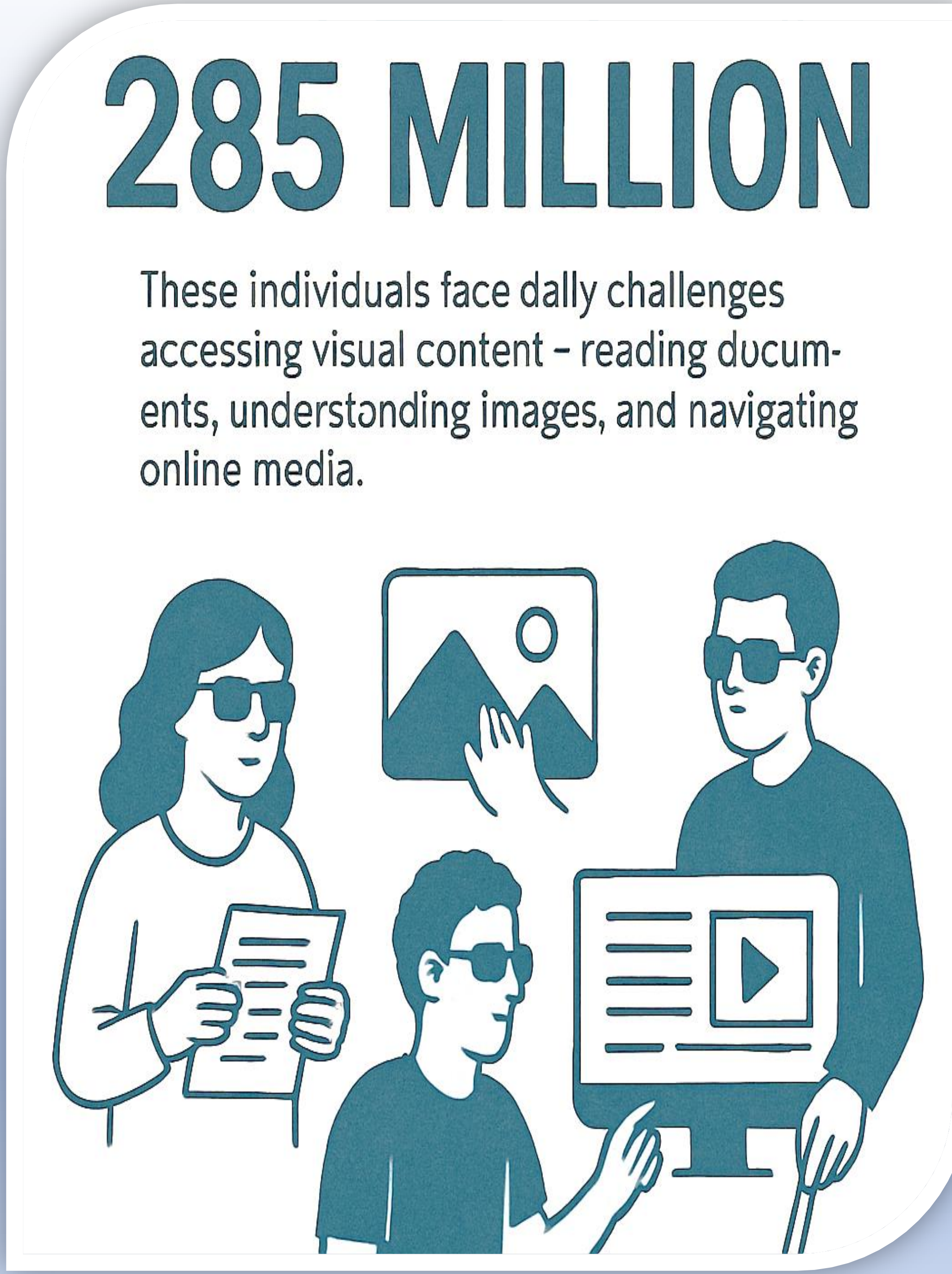
Introduction & Motivation Context

- **<u>Current Challenges:</u>**
  - Limited screen readers
  - Low-quality or robotic audio
  - Inaccessibility of images, charts, and scanned documents
  - Inability to understand visual content (charts, signs, forms)
  - No modular assistive tools using modern AI
  - Limited availability of affordable, real-time assistive tools

- **<u>Potential Solution</u>**:
- A multi-agent AI system that:
  - ✔ Accepts any document/image
  - ✔ Understands content using Gemini API
  - ✔ Converts it into natural speech using Murf API
  - ✔ Outputs real-time, accessible audio



Problem Statement

➢ <u>Problem Statement:</u>
- Visually impaired individuals struggle to independently access written or visual information in physical and digital formats.

➢ <u>Objective:</u>
- To design a modular, scalable AI system that empowers blind users with voice-based understanding of documents and images.

➢ <u>Specific Goals:</u>
- Build multi-agent interaction
- Use advanced APIs (Gemini & Murf)
- Provide regional language support (future goal)
- Ensure real-time, accessible experience

Sustainable Development Goal(SDG)

➢ Goal Chosen:

- Reduced Inequalities (Primary)
- Also supports SDG 4 (Quality Education) and
- SDG 3(Well-being)

➢ Problem statement :

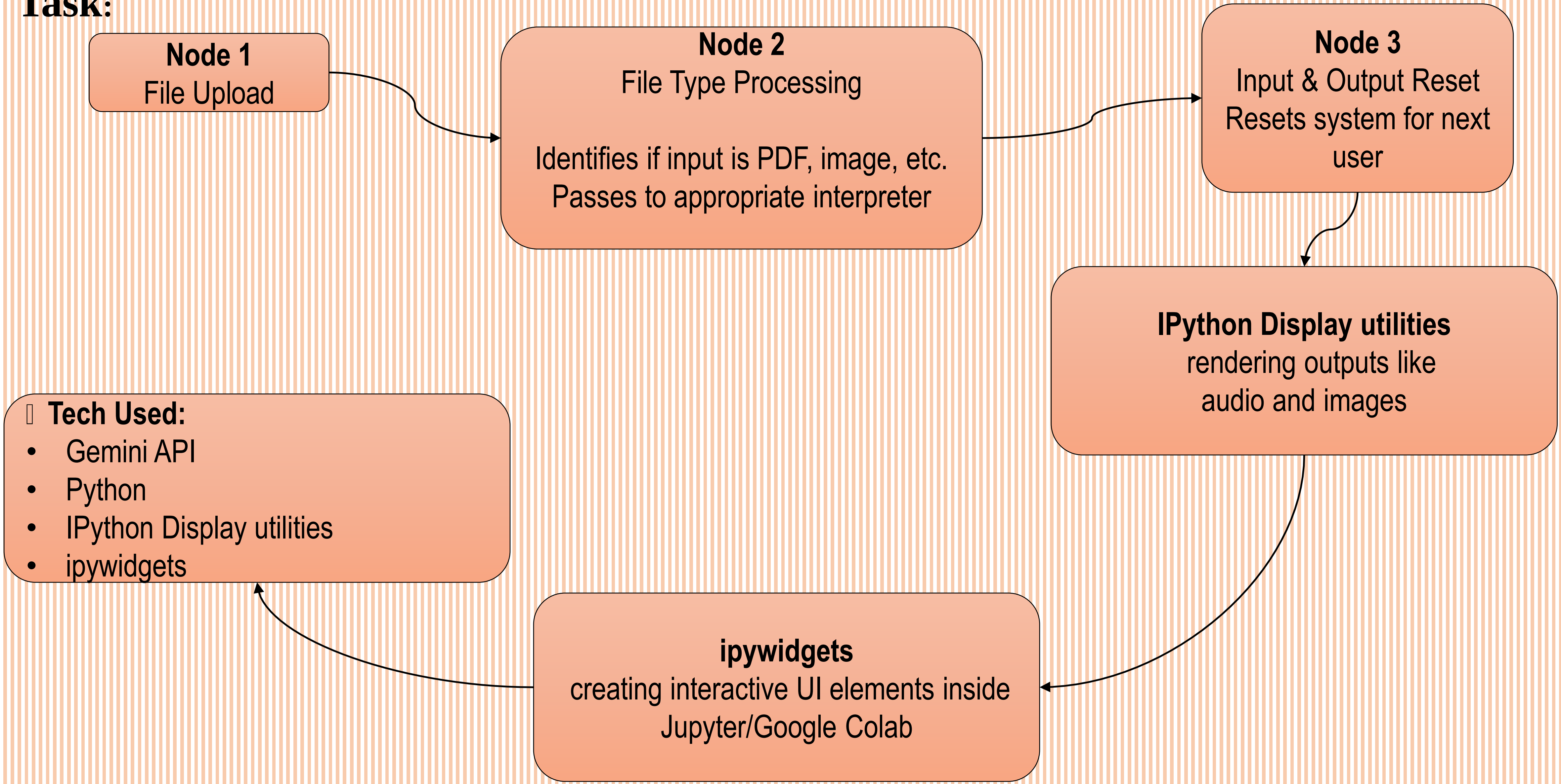
- Millions of blind people face barriers in accessing information. VocalEyes reduces inequality by enabling autonomous access to any content via voice."

 Rationale :

- Creating inclusive technology is critical to achieving equity in education, employment, and access to services.

Agent 1 – Document/Image Understanding

Task:



Agent 2 – Voice Response Generator

Task:

1. Text Input from Agent 1
2. Send to Murf API
3. Generate Audio URL
4. Stream/playback audio to user

□ Output:

- Natural-sounding speech in English (expandable to regional languages)

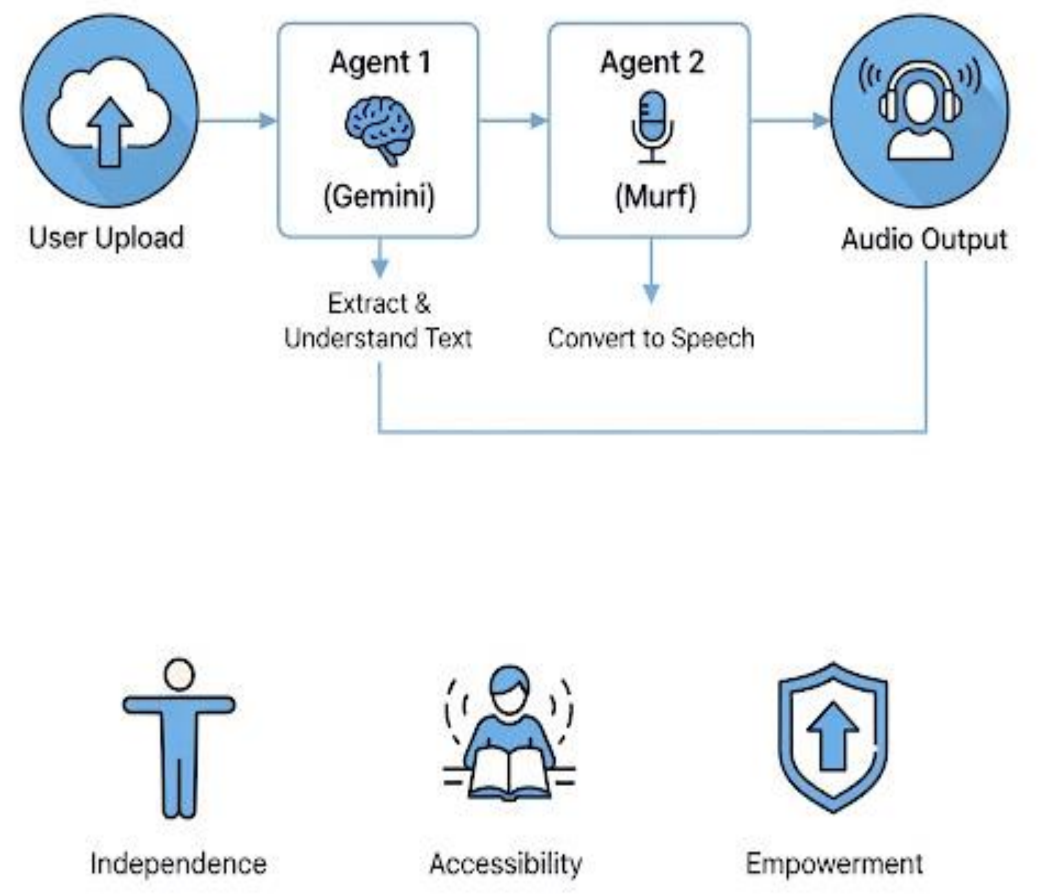
□ Tech Used:

- Murf API
- Python
- Web Audio streaming

Multi-Agent Workflow Diagram

➢ **Workflow Summary:**

→ User uploads file
→ Agent 1 (Gemini) extracts/understands text
→ Text passed to Agent 2
→ Agent 2 uses Murf to convert to speech
→ Audio is streamed to the user



 Result:

- Independence, accessibility and empowerment

Tools & Resources

Tool / API	Usage
Gemini API	Image/doc text extraction, content analysis
Murf API	High-quality text-to-speech
Python	Backend logic, API integration
Ipywidgets	For creating interactive UI elements
IPython Display utilities	For rendering outputs like audio and images.
GitHub	Project version control

Impact & Effectiveness

- **Reach & Inclusivity:**
  - 285+ million visually impaired globally useful in education, employment, government, health

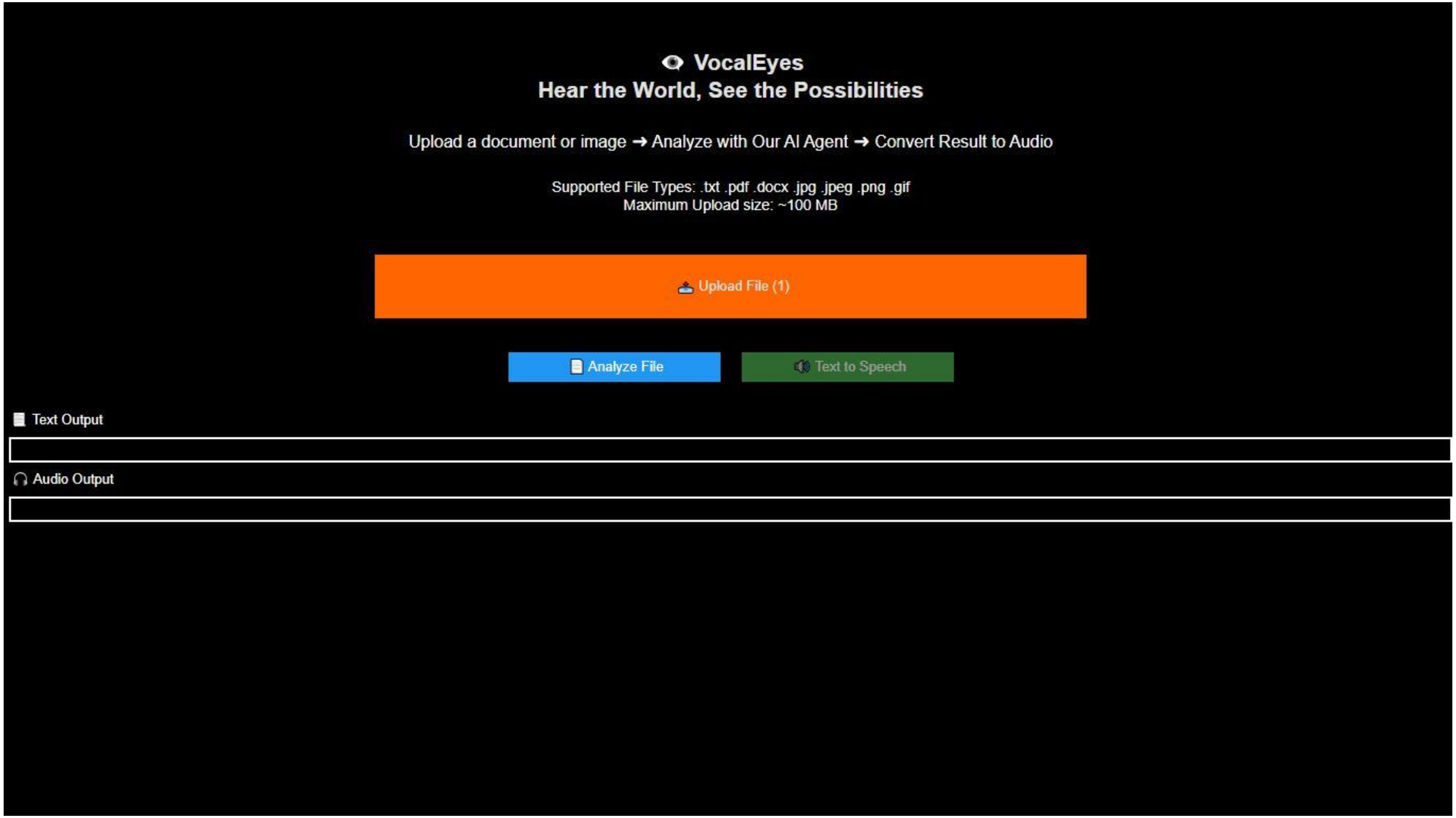
- **Impact Areas:**
  - Access to learning material
  - Reading official forms & reports
  - Understanding daily documents without help

- **AI Agent Benefits:**
  - Scalable, plug-and-play
  - Easy to extend with more APIs
  - High performance with minimal user effort

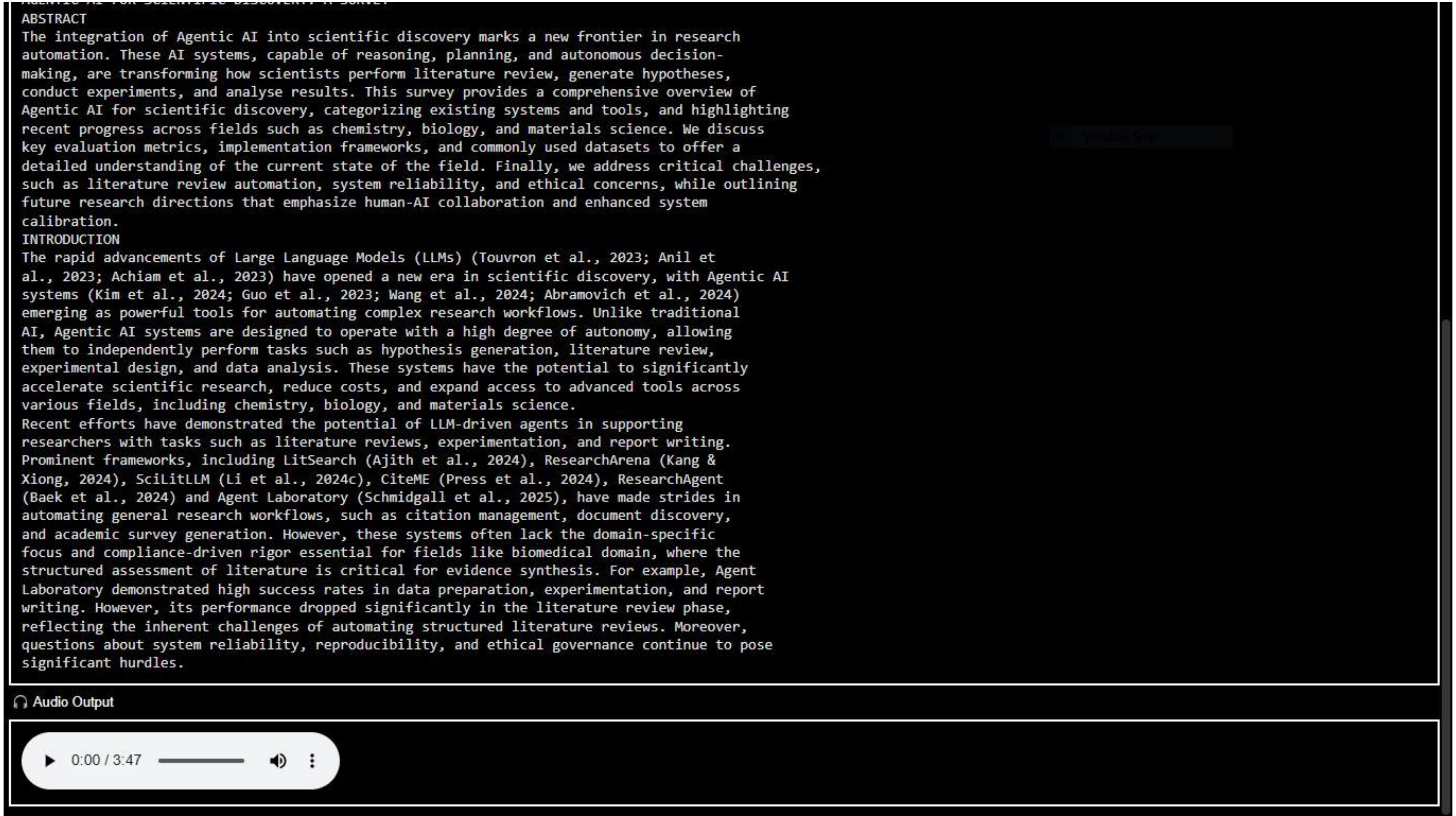
Why It Will Work

- ✓ Uses cutting-edge APIs (Gemini + Murf)
- ✓ Multi-agent = scalable, maintainable
- ✓ Simple, intuitive UI
- ✓ Real-world need with clear SDG alignment
- ✓ Cloud-based = easy deployment
- ✓ Natural voice output = high user satisfaction
- ✓ Extensible: can add Braille output or translation

Screenshots / Results



Screenshots / Results(.pdf file extraction)



Screenshots / Results(.jpeg file analysis)

Upload File (1)

Analyze File

Text to Speech

Text Output

Processing `Agentic_AI_Workflow.JPG` (type: image/jpeg)

The image contains two figures illustrating the application of Agentic AI in scientific discovery.

Figure 1: Agentic AI workflow for scientific discovery depicts a circular workflow diagram. The diagram shows a series of steps, each represented by a circular image containing a cartoon robot performing a task, and connected by arrows indicating the flow of the process. The steps are:

1. **Idea Generation & Literature Review:** Shows a robot reviewing documents.
2. **Research Planning & Experiment Design:** Shows a robot analyzing a graph.
3. **Data Preparation & Experiment Execution:** Shows a robot in a lab setting.
4. **Results Analysis:** Shows a robot analyzing data on a computer screen.
5. **Report Writing & Synthesis:** Shows a robot writing a report.
6. **Paper Review:** Shows a robot reviewing a document.

The circular nature emphasizes the iterative nature of the scientific process. The use of cartoon robots emphasizes the automation aspect of Agentic AI.

Figure 2: AI Agents frameworks for scientific discovery presents a tree-like diagram categorizing different AI agent frameworks across various scientific disciplines. The main branches represent scientific fields: Chemistry, Biology, Materials Science, General Science, and Machine Learning. Each branch further subdivides into specific AI agent tools and projects, listed with their names and sometimes citations. This figure provides a taxonomy of available AI tools for different scientific research areas.

In summary, the image visually explains both the process (Figure 1) and the available tools (Figure 2) of using Agentic AI in scientific research. The use of clear and concise visuals, along with text labels, makes the information easily digestible.

Audio Output

0:00 / 2:05

Project Links

- Google Colab Link: Click Here (Recommended)
- Github Repo Link: Click Here

- **Important:** Our project generates audio output which is not possible to showcase in a PPT. So, the generated output audio files and provided input files are uploaded Here in Google Drive.

- Note: Please Get APIs in order to run the code.

Google Gemini API: https://aistudio.google.com/apikey

Murf API key: https://murf.ai/api/docs/text-to-speech/overview

Conclusion & Future Scope

Conclusion:

- **VocalEyes** bridges the accessibility gap for the visually impaired using intelligent AI agents.
- It's a step toward a more inclusive, voice-powered digital world.

Future Scope:

- Support for regional languages
- Offline version for remote use
- Integrate object recognition and scene narration
- Add Braille reader compatibility
- Government-level deployment with NGO support

Thank You!